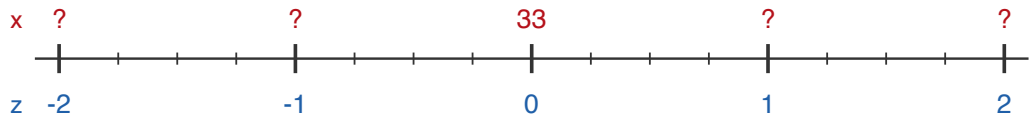
Z-Values

The mean is $\mu = 33$ and the standard deviation is $\sigma = 4$.

The z-value of a data value x is

$$z = \frac{x - \mu}{\sigma}$$

a) Fill in the x-values for each z-value on the number line.

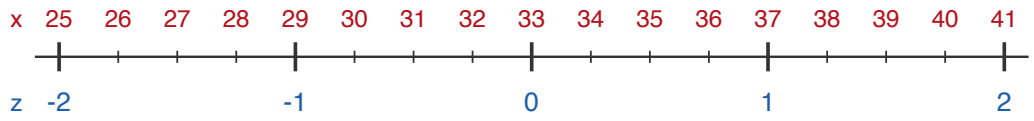


For each data value below:

- **Visually:** mark x on the number line and read off its z-value.
- **Calculate:** find the z-value with the formula.

Check that your two answers match.

b) $x = 39$



Visually: $z =$ _____

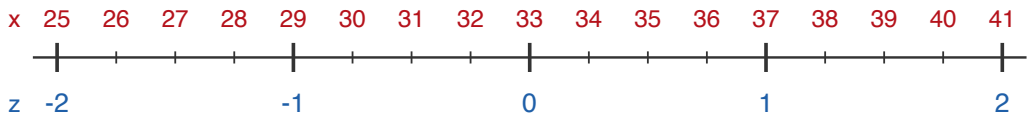
Calculate: $z = \frac{39 - 33}{4} =$ _____

c) $x = 31$

Visually: $z =$ _____

Calculate: $z =$ _____

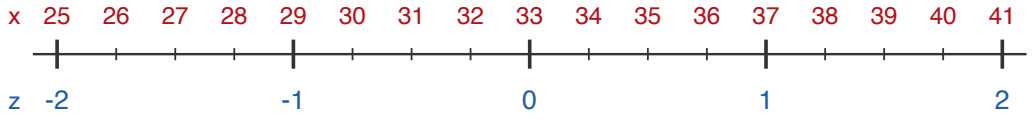
d) $x = 26$



Visually: $z =$ _____

Calculate: $z =$ _____

e) $x = 36$



Visually: $z =$ _____

Calculate: $z =$ _____